

Question Bank for Module-2 : Fourier Series

Type - I

1. If $f(x) = \begin{cases} a, & 0 < x < \pi \\ -a, & \pi < x < 2\pi \end{cases}$ Prove that: $f(x) = \frac{4a}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$
2. Obtain Fourier series for $f(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi \\ 0, & \pi \leq x \leq 2\pi \end{cases}$
Hence, deduce that $\frac{1}{2} = \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots$ or $1 = \sum_{n=1}^{\infty} \frac{2}{(2n-1)(2n+1)}$
3. Obtain Fourier series for $f(x) = \begin{cases} 1, & 0 \leq x \leq \pi \\ 2 - \frac{x}{\pi}, & \pi \leq x \leq 2\pi \end{cases}$
4. Obtain Fourier series for $f(x) = \begin{cases} -\pi, & 0 \leq x \leq \pi \\ x - \pi, & \pi \leq x \leq 2\pi \end{cases}$
state the value of the series at $x = \pi$ and hence, show that $\frac{\pi^2}{8} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$
5. Obtain the Fourier expansion of $f(x) = \left(\frac{\pi-x}{2} \right)^2$ in the interval $0 \leq x \leq 2\pi$ and $f(x+2\pi) = f(x)$
Also deduce that

(i) $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$	(ii) $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$
(iii) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$	(iv) $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots$
6. Find the Fourier Series of the function $f(x) = e^{-x}$, $0 < x < 2\pi$ and $f(x+2\pi) = f(x)$
Hence, deduce the value of $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^2+1}$ Further derive the series for cosech π

Type-II

1. If $f(x) = \begin{cases} 2, & 0 \leq x \leq 2 \\ x, & 2 \leq x \leq 4. \end{cases}$ Find the Fourier series for $f(x)$.
2. Find Fourier series for $f(x)$ which is defined as: $f(x) = \begin{cases} \pi x, & 0 < x < 1 \\ 0, & x = 1 \\ \pi(x-2), & 1 < x < 2 \end{cases}$
3. Obtain Fourier series for the function $f(x) = 2x - 1$ in $0 < x < 3$
4. If $f(x) = \frac{L}{2} - x$, $0 \leq x \leq L$. Find the Fourier series for $f(x)$
5. Expand $f(x) = \begin{cases} l - x, & 0 < x < l \\ 0, & l < x < 2l \end{cases}$ into a Fourier series.
6. Find Fourier series for $f(x) = 2 - \frac{x^2}{2}$ in $0 \leq x \leq 2$
7. Find Fourier series for $f(x) = \begin{cases} -x, & 0 < x < l \\ 0, & l < x < 2l \end{cases}$ Hence, deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$
8. Find the Fourier expansion of $f(x) = 2x - x^2$, $0 \leq x \leq 3$ whose period is 3. Also plot the graph of $f(x)$

TYPE-III

1. Obtain Fourier series in the interval $(-\pi, \pi)$ for the function $f(x)$ given by $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \sin x, & 0 < x < \pi \end{cases}$
Hence deduce that (i) $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots = \frac{1}{2}$ (ii) $\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots = \frac{\pi-2}{4}$

2. Find the Fourier series of the function defined as $f(x) = \begin{cases} x + \pi & \text{for } 0 \leq x \leq \pi \\ -x - \pi & \text{for } -\pi \leq x \leq 0 \end{cases}$
3. Find the Fourier series for $f(x)$ where $f(x) = \begin{cases} \frac{1}{2}, & -\pi < x < 0 \\ \frac{x}{\pi}, & 0 < x < \pi \end{cases}$ Hence deduce that $\frac{\pi^2}{8} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$
4. Determine the Fourier expansion for the function given by $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$
5. Obtain Fourier series for $f(x) = x - x^2$, in the range $(-\pi, \pi)$. Hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$
6. Obtain a Fourier series for $f(x) = |\cos x|$ in the range $-\pi < x < \pi$
7. Obtain Fourier series for $f(x)$ where $f(x) = \begin{cases} x + \frac{\pi}{2} & -\pi < x < 0 \\ \frac{\pi}{2} - x & 0 < x < \pi \end{cases}$
Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} - \dots$
Also deduce that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$
8. For the interval $-\pi \leq x \leq \pi$, prove that $\frac{\pi^2}{12} - \frac{x^2}{4} = \cos x - \frac{1}{4}\cos 2x + \frac{1}{9}\cos 3x - \frac{1}{16}\cos 4x + \dots$
9. Determine the Fourier expansion for the function given by $f(x) = \pi^2 - x^2$, $-\pi \leq x \leq \pi$
10. Prove that for $-\pi < x < \pi$, $\frac{x(\pi^2 - x^2)}{12} = \frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \dots$

TYPE-IV

1. If $f(x) = 9 - x^2$ for $-3 < x < 3$, obtain Fourier series of $f(x)$
2. A periodic function of period 4 is defined as $f(x) = |x|$ in $-2 < x < 2$. Find its Fourier series expansion.
Hence, deduce that $\sum \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$
3. Determine the Fourier expansion of the function given by $f(x) = \begin{cases} 0, & -3 < x < -1 \\ 1 + \cos \pi x, & -1 < x < 1 \\ 0, & 1 < x < 3 \end{cases}$
4. Find Fourier series for $f(x)$ defined by $f(x) = \begin{cases} a(x-l), & -l < x < 0 \\ a(x+l), & 0 < x < l \end{cases}$
Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$
5. Obtain the Fourier series $f(x) = a^2 - x^2$ in $(-a, a)$ $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$
6. Find the Fourier series for $f(x) = x - x^3$ in the interval $-1 < x < 1$.

TYPE -V

1. Find half range cosine series for $f(x) = \begin{cases} 1, & 0 < x < a/2 \\ -1, & a/2 < x < a \end{cases}$
2. Find the half range Fourier sine expansion for $f(x) = x^2$ in the interval $0 \leq x \leq 3$.

3. Obtain the Fourier cosine expansion of the periodic function defined by $f(x) = \sin\left(\frac{\pi x}{l}\right), 0 < x < l$

Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$

4. Find half – range cosine series for $f(x) = (x - 1)^2$ in $0 < x < 1$

Hence deduce that (i) $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ (ii) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$ (iii) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

5. Obtain the Fourier expansion of $f(x) = x \sin x$ as a cosine series in $(0, \pi)$ and show that

$$1 + \frac{2}{1.3} - \frac{2}{3.5} + \frac{2}{5.7} - \dots = \frac{\pi}{2}$$

6. Obtain a half – range cosine series for $f(x)$, where $f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 < x < 2 \end{cases}$

Hence deduce that $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$

7. Find half range Fourier sine series of $f(x)$ defined as $f(x) = \begin{cases} kx, & 0 < x < \frac{L}{2} \\ k(L - x), & \frac{L}{2} < x < L \end{cases}$

Hence deduce that (i) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$ (ii) $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$

8. Obtain the half – range sine series for $f(x) = \pi x - x^2$ in $(0, \pi)$ and hence deduce that

(i) $\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \frac{1}{7^6} + \dots = \frac{\pi^6}{960}$ (ii) $\frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \frac{1}{4^6} + \dots = \frac{\pi^6}{945}$

9. If $f(x) = \begin{cases} \sin x & \text{for } 0 \leq x \leq \pi/4 \\ \cos x & \text{for } \pi/4 \leq x \leq \pi/2, \end{cases}$ expand $f(x)$ in a series of sines.

10. Find half range sine series of period $2l$ for $f(x) = \begin{cases} \frac{2x}{l} & \text{for } 0 \leq x \leq \frac{l}{2} \\ \frac{2(l-x)}{l} & \text{for } \frac{l}{2} \leq x \leq l \end{cases}$